ENTERPRISE ARCHITECTURE BLUEPRINT

Cloud Accounting & Business Management Platform

(Codename "Ledger" - Zoho Books-class product)

VOLUME 9 OF 9 - FINAL VOLUME

UI/UX DESIGN SPECIFICATION & API DESIGN DOCUMENT

Design System - Screen Specs - Endpoint Catalog - Webhooks - Error Codes - Blueprint Closure

Field	Value
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Upstream Dependency	Vol. 1 Personas & Value Streams; Vol. 2 Resource Ownership Map; Vol. 3 React/Component Library Decision & API Conventions; Vol. 5 Webhook Platform; Vol. 6 Auth Model
Status	Completes the 9-volume Enterprise Architecture Blueprint

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1. Introduction & Scope

1.1 Purpose

This is the ninth and final volume of the Enterprise Architecture Blueprint. It has two parts. Part A is the UI/UX Design Specification: the design system, information architecture, and screen-level specifications for the six value streams defined in Vol. 1 Section 7, built on the React/component-library decision from Vol. 3 Section 2.3. Part B is the API Design Document: the full endpoint catalog expanding Vol. 2 Section 11's resource-ownership map, request/response examples, the webhook payload catalog expanding Vol. 5 Section 7.1, the error code reference formalizing Vol. 3 Section 4.3, and versioning policy. Section 14 closes the entire blueprint with a full nine-volume traceability index.

1.2 Why UI/UX and API Design Share One Volume

The UI is explicitly a client of the same API every third-party integrator uses (Vol. 2 Principle, Vol. 3 Section 5) - there is no separate internal API. Documenting them together keeps that principle enforced by construction: every screen specification in Part A cites the exact endpoint(s) in Part B it calls, so the two documents cannot drift apart the way a UI spec and an API spec maintained separately often do.

PART A: UI/UX DESIGN SPECIFICATION

2. Design System Foundations

2.1 Design Tokens

Token Category	Values (representative)	Notes
Color - Primary	Navy (#1F3864) for brand/headers, Accent Blue (#2E75B6) for interactive elements	Matches this blueprint's own document styling for internal consistency
Color - Semantic	Success (green), Warning (amber), Error (red), Info (blue)	Used consistently for invoice/bill status badges (Vol.2 Section 8.1 state machine colors)
Typography	System font stack for performance; a distinct monospace for financial figures to aid column alignment	Financial amounts always right-aligned, tabular-figure font variant
Spacing scale	4px base unit (4/8/12/16/24/32/48/64)	Consistent density across dense financial tables and lighter marketing-style portal pages
Elevation/shadow	3-tier system: flat, raised (cards), overlay (modals/dropdowns)	Keeps modal/dropdown hierarchy visually unambiguous in dense screens like the invoice editor

2.2 Component Library

Built on the Vol. 3 Section 2.3 decision (React + a headless UI kit such as Radix primitives). Core components: data table (with sort/filter/pagination matching Vol. 3 Section 5's cursor-pagination convention), form inputs with inline validation (surfacing the namespaced error codes from Section 11), status badge, money input (locale-aware, Vol. 1 BC-12), date range picker, modal/drawer, toast/notification, and the recurring approval-workflow widget (Vol. 1 BC-16).

2.3 Design-to-Dev Handoff

- Every screen spec (Section 4) ships with a component inventory referencing this section's library - no ad hoc one-off components without an explicit exception noted.
- Design tokens (Section 2.1) are implemented as a shared theme file consumed by both the web app and portal app (Vol.3 Section 3.2 /libs/shared/ui-components), never hand-copied per screen.
- Accessibility annotations (Section 5) are part of the handoff artifact, not a separate post-hoc audit.

3. Information Architecture

3.1 Primary Navigation (Web App)

Top-level navigation is organized by bounded context (Vol. 2 Section 3.2), matching how permissions are granted (Vol. 6 Section 3.3) so a user's nav only ever shows sections they can act on - there is no navigation item that leads to a permission-denied screen.

Nav Section	Bounded Context	Visible To (Vol.6 Section 3.3 Roles)
Dashboard	Reporting & Analytics (CTX-11)	All roles (scoped to their own data visibility)
Sales	Receivables (CTX-03)	Owner/Admin, Accountant, Sales Staff
Purchases	Payables (CTX-04)	Owner/Admin, Accountant, Purchasing Staff
Banking	Banking (CTX-05)	Owner/Admin, Accountant
Inventory	Inventory (CTX-07)	Owner/Admin, Inventory Manager, Sales/Purchasing Staff (read-only)
Projects	Projects (CTX-08)	Owner/Admin, Project Manager
Reports	Reporting & Analytics (CTX-11)	All roles (report set varies by role)
Settings	Identity & Tenant, Automation (CTX-01, CTX-14)	Owner/Admin only for org/user/role settings

3.2 Portal Navigation (Customer/Vendor)

A deliberately minimal, separate navigation (Vol. 6 Section 3.1 portal auth) - customers see Invoices, Payments, and Messages; vendors see Purchase Orders, Bills, and Messages. Neither portal exposes any internal-staff navigation item, consistent with the portal's restricted API scope (Vol. 2 Section 11).

4. Screen Specifications by Value Stream

Each value stream from Vol. 1 Section 7 gets its key screens specified here at a wireframe-description level of detail (not pixel-perfect mockups, which belong to a dedicated design tool) - enough for engineering to build against and for the API cross-references (Part B) to be concrete.

4.1 VS-01 Order-to-Cash Screens

Screen	Key Elements	Calls (Part B)
Quote/Invoice List	Data table: customer, number, status badge, amount, due date; filter by status; bulk actions (send reminder)	GET /v1/invoices
Invoice Editor	Customer selector, line-item table with item/qty/price/tax, totals summary, save-as-draft vs send actions	POST /v1/invoices, PATCH /v1/invoices/{id}
Payment Recording	Amount, method, date, reference number; partial-payment support with running balance display	POST /v1/invoices/{id}/payments
Customer Portal - Invoice View	Read-only invoice detail, Pay Now button (Vol.5 Section 4.3 payment link), comment thread	GET /v1/portal/invoices/{id}, POST /v1/portal/invoices/{id}/pay

4.2 VS-02 Procure-to-Pay Screens

Screen	Key Elements	Calls (Part B)
Purchase Order Editor	Vendor selector, line items, expected delivery date, approval-status indicator (Vol.1 BC-03)	POST /v1/purchase-orders
Bill Entry / 3-Way Match Review	Side-by-side PO vs Receipt vs Bill comparison, variance highlighting (Vol.2 Section 9.3), approve/flag actions	GET /v1/bills/{id}/match-status, POST /v1/bills/{id}/approve
Vendor Portal - Bill Submission	Upload/submit bill against a received PO, status tracker	POST /v1/portal/bills

4.3 VS-03 Record-to-Report & VS-06 Close-the-Books Screens

Screen	Key Elements	Calls (Part B)
Chart of Accounts	Hierarchical account tree, add/edit account, system-defined accounts locked from deletion (Vol.4 Section 4.1)	GET /v1/accounts
Journal Entry Editor	Multi-line debit/credit grid with running balance-check indicator (must net to zero before save)	POST /v1/journal-entries
Period Close Wizard	Checklist UI walking through reconciliation status, outstanding approvals, then period lock confirmation (Vol.1 VS-06)	GET /v1/fiscal-periods/{id}/close-readiness, POST /v1/fiscal-periods/{id}/lock
Financial Statements	P&L, Balance Sheet, Cash Flow with period comparison selector	GET /v1/reports/profit-and-loss, etc.

5. Responsive & Accessibility Standards

Standard	Requirement
Responsive breakpoints	Desktop-first for the core web app (dense financial tables genuinely need width); portal app is mobile-first (Vol.1 PER-09/PER-10 are more likely on mobile)
WCAG conformance target	WCAG 2.1 AA across both web app and portal - color contrast, keyboard navigation, screen-reader labeling on every form input and status badge
Keyboard operability	Every action reachable via keyboard alone, critical for the dense data-entry screens (invoice/bill editors, Section 4)
Status communication	Status is never color-only (Vol.2 Section 8.1 state machine) - every status badge pairs color with a text label and, where relevant, an icon
Error messaging	Every namespaced error code (Section 11) maps to a human-readable, actionable message - never surfacing a raw error code to an end user

PART B: API DESIGN DOCUMENT

6. API Conventions (Recap & Expansion of Vol. 3 Section 5)

Convention	Detail
Base URL	https://api.ledgerplatform.example/v1
Versioning	URL-path versioned (Vol.3 Section 5); this document specifies v1 fully
Auth	Bearer JWT (Vol.6 Section 3.2); portal endpoints use a separate portal-scoped token
Pagination	Cursor-based: ?cursor=...&limit=50, response includes nextCursor
Idempotency	Idempotency-Key header required on all financial-mutating POSTs (Vol.3 Section 5, Vol.5 Principle I4)
Content type	application/json exclusively; file uploads (attachments, Vol.1 BC-14) use multipart/form-data on dedicated endpoints

7. Endpoint Catalog by Bounded Context

Expands Vol. 2 Section 11's resource-ownership table into concrete methods and paths. This is not exhaustive of every field but fixes the full surface area every module must implement.

7.1 Identity & Tenant (CTX-01)

Method	Path	Purpose
POST	/v1/auth/login	Authenticate, returns access + refresh token (Vol.6 Section 3.1)
POST	/v1/auth/refresh	Exchange refresh token for a new access token
POST	/v1/auth/mfa/verify	Complete MFA challenge (Vol.6 Section 3.1)
GET/POST	/v1/users	List / invite tenant users
GET/POST	/v1/roles	List / create custom roles (Vol.6 Section 4.2 ceiling rule enforced server-side)

7.2 Ledger (CTX-02)

Method	Path	Purpose
GET	/v1/accounts	List chart of accounts
POST	/v1/accounts	Create an account
POST	/v1/journal-entries	Create a manual journal entry (Vol.4 Section 4.1 chk_one_sided enforced)
GET	/v1/journal-entries/{id}	Fetch a journal entry with its lines
POST	/v1/fiscal-periods/{id}/lock	Lock a fiscal period (Vol.1 VS-06)

7.3 Receivables (CTX-03)

Method	Path	Purpose
GET/POST	/v1/quotes	List / create quotes
POST	/v1/quotes/{id}/convert	Convert quote to sales order (Vol.1 VS-01)
GET/POST	/v1/invoices	List / create invoices
POST	/v1/invoices/{id}/send	Send invoice (triggers Vol.2 Section 9.1 flow)
POST	/v1/invoices/{id}/payments	Record a payment against an invoice
POST	/v1/invoices/{id}/void	Void an invoice (append-only reversal, Vol.4 Section 2 D2/D6)

7.4 Payables (CTX-04) & Inventory (CTX-07)

Method	Path	Purpose
GET/POST	/v1/purchase-orders	List / create purchase orders
GET/POST	/v1/bills	List / create vendor bills
POST	/v1/bills/{id}/approve	Approve a bill post-3-way-match (Vol.2 Section 9.3)
GET/POST	/v1/items	List / create inventory items
POST	/v1/items/{id}/adjustments	Record a stock adjustment

7.5 Banking (CTX-05) & Payments (CTX-06)

Method	Path	Purpose
GET	/v1/bank-accounts	List connected bank accounts (Vol.5 Section 3.2)
GET	/v1/bank-transactions	List imported bank transactions
POST	/v1/bank-transactions/{id}/reconcile	Match a transaction to an invoice/bill payment
POST	/v1/payment-links	Create a payment link for an invoice (Vol.5 Section 4.3)

7.6 Reporting (CTX-11)

Method	Path	Purpose
GET	/v1/reports/profit-and-loss	Generate P&L for a date range
GET	/v1/reports/balance-sheet	Generate balance sheet as of a date
GET	/v1/reports/custom	Run a saved custom report (Vol.1 BC-13)

8. Request/Response Examples

8.1 Create Invoice

POST /v1/invoices

```
Request:
{
  "customerId": "cus_7f21a",
  "currency": "USD",
  "dueDate": "2026-08-15",
  "lineItems": [
    { "itemId": "itm_44a1", "quantity": 2, "unitPrice": 250.00 }
  ]
}
Headers: Idempotency-Key: 5c1e-...

Response (201):
{
  "id": "inv_0042",
  "invoiceNumber": "INV-0042",
  "status": "Draft",
  "totalAmount": "500.0000",
  "balanceDue": "500.0000",
  "currency": "USD"
}
```

8.2 Record Payment

POST /v1/invoices/inv_0042/payments

```
Request:
{ "amount": 500.00, "method": "bank_transfer", "paymentDate": "2026-08-01" }
Headers: Idempotency-Key: 9a2f-...

Response (201):
{
  "id": "pay_1188",
  "invoiceId": "inv_0042",
  "amount": "500.0000",
  "invoiceStatus": "Paid"
}
```

9. Webhook Payload Catalog (Expands Vol. 5 Section 7.1)

Webhook Event	Payload Fields (representative)	Signed
invoice.sent	invoiceId, invoiceNumber, customerId, totalAmount, currency, sentAt	Yes (HMAC-SHA256, Vol.5 Section 7.1)
invoice.paid	invoiceId, paymentId, amount, balanceDue, paidAt	Yes
bill.approved	billId, vendorId, totalAmount, approvedAt	Yes
payment.failed	paymentId, invoiceId, failureReason, failedAt	Yes
journal.posted	journalEntryId, accountIds, totalDebit, totalCredit, postedAt	Yes

10. Error Code Reference (Formalizing Vol. 3 Section 4.3)

Error Code	HTTP Status	Meaning
RECEIVABLES.INVOICE_ALREADY_PAID	409	Attempted to void/edit an invoice that is already fully paid
RECEIVABLES.INSUFFICIENT_STOCK	422	Invoice/sales order line item exceeds available inventory (Vol.2 Section 9.1)
LEDGER.PERIOD_LOCKED	409	Attempted to post to a locked fiscal period (Vol.1 VS-06)
LEDGER.UNBALANCED_ENTRY	422	Journal entry debits and credits do not match (Vol.4 Section 4.1 chk_one_sided)
PAYABLES.MATCH_VARIANCE	422	3-way match found a quantity or price variance requiring manual review (Vol.2 Section 9.3)
AUTH.MFA_REQUIRED	403	Action requires MFA verification (Vol.6 Section 3.1) not yet completed in this session
COMMON.IDEMPOTENCY_CONFLICT	409	Idempotency-Key reused with a different request body
COMMON.RATE_LIMITED	429	Tenant/API-key rate limit exceeded (Vol.3 Section 7.1, Vol.5 Section 7.2)

11. API Versioning & Deprecation Policy

- A new major version (/v2) is introduced only for breaking changes; additive fields and new optional parameters never require a version bump (Vol.3 Section 5).
- A deprecated version is supported for a minimum of 12 months after its successor ships, with deprecation notices returned via a response header on every call.
- Webhook payload schemas (Section 9) follow the same additive-only rule as internal domain events (Vol.3 Section 4.2 CS-06); a breaking payload change ships as a new event name, not a mutated existing one, so existing integrator subscriptions never silently break.

12. Developer Portal & SDK Strategy

A public developer portal (documentation, API key management, webhook subscription management, Vol.5 Section 7.1) is a Release 2/3 capability. At Release 1, API documentation is published as a static OpenAPI-generated reference site, and API keys are issued manually by the platform team on request - sufficient for early integrator partners without building a self-service portal before there is integrator demand to justify it.

13. Full Blueprint Closure: Nine-Volume Traceability Index

This is the capstone index for the entire Enterprise Architecture Blueprint. It maps every volume to what it produced and what still depends on it, so the blueprint can be read as a single coherent system rather than nine separate documents.

Vol.	Volume Title	Core Artifact Produced	Consumed By
1	Business Architecture	21 capability domains (BC-01…BC-21), 12 personas, 6 value streams, module scope catalog	All subsequent volumes
2	Application Architecture	16 bounded contexts (CTX-01…CTX-16), C4 model, module dependency graph, event catalog	Vol. 3-9
3	Technical Architecture	Stack decisions (NestJS/React/Postgres/Kafka/Redis/OpenSearch), folder structure, coding standards, logging/caching/queueing, multi-tenancy strategy	Vol. 4-9
4	Data Architecture	ER diagrams, physical schema (DDL), RLS implementation, indexing/partitioning, migration & retention strategy	Vol. 5-9
5	Integration Architecture	Banking/payment/tax/e-commerce integration design, resilience patterns, public webhook platform	Vol. 6-9
6	Security Architecture	Threat model, IAM/RBAC, encryption/secrets, defense-in-depth, audit trail, compliance mapping	Vol. 7-9
7	Infrastructure & Deployment	Cloud/managed-service mapping, deployment topology, CI/CD, scaling, observability, DR	Vol. 8-9
8	Testing & QA Specification	Test pyramid, security/performance/resilience test plans, UAT acceptance criteria, release gates	Vol. 9, engineering execution
9	UI/UX & API Design (this volume)	Design system, screen specs, full endpoint catalog, webhook payloads, error codes	Engineering execution

14. Blueprint Completion Summary & Recommended Next Steps

14.1 What This Blueprint Delivers

Nine volumes now define, at enterprise depth, everything a Zoho Books-class platform needs before development begins: what the business does and for whom (Vol.1), how that decomposes into software boundaries (Vol.2), what it is built with and how it is coded (Vol.3), how data is modeled and protected in the database (Vol.4), how it talks to the outside financial world (Vol.5), how it is secured (Vol.6), how it runs in production (Vol.7), how it is proven correct before every release (Vol.8), and what it looks like and how it is called (Vol.9).

14.2 Recommended Next Steps

1. Circulate all nine volumes for stakeholder sign-off, prioritizing the open questions flagged at the end of each volume (target market countries, cloud provider confirmation, QA staffing, SLA commitments) since several later decisions (Vol.5, Vol.7) are contingent on those answers.
2. Stand up the monorepo (Vol.3 Section 3) and base infrastructure (Vol.7 Section 6 Terraform modules) as the first engineering sprint, before any feature module work begins, so every subsequent module is built against the correct scaffolding from day one.
3. Sequence Release 1 (MVP) engineering work by the module priority already fixed in Vol.1 Section 6, respecting the layered dependency order in Vol.2 Section 7 (build Identity & Tenant and Ledger first, since every other module depends on them).
4. Treat this blueprint as a living set of documents: when a real implementation decision deviates from what is written here, update the relevant volume rather than letting the documentation and the system silently drift apart - the whole value of this exercise depends on that discipline.

14.3 Closing Note

This concludes the nine-volume Enterprise Architecture Blueprint. Thank you for the opportunity to build this out at the depth you asked for - a genuine engineering foundation rather than a template SRS.